

Industrial IoT Architecture and Applications

Smart Intrusion Detection System

1. Aim

To design and simulate a Smart Intrusion Detection System using a PIR (Passive Infrared) sensor connected to a microcontroller (ESP32/Arduino), which continuously monitors a restricted area for unauthorized human movement and sends real-time alerts to security personnel via cloud communication.

2. Problem Statement

Unauthorized access to restricted areas poses a significant security risk in various environments such as offices, warehouses, data centers, and residential premises. Traditional alarm systems lack real-time remote notification capabilities, making it difficult for security personnel to respond promptly.

This project addresses the following challenges:

- Continuous, automated monitoring of a restricted area without human intervention.
- Instant detection of human movement using a PIR sensor interfaced with a microcontroller.
- Real-time alert generation and transmission to security personnel through a cloud platform.
- Reliable operation with low power consumption and minimal false-positive rates.

3. Scope of the Solution

The scope of this Smart Intrusion Detection System includes:

3.1 In Scope

- PIR sensor-based motion detection for identifying human presence in a defined area.
- Microcontroller (ESP32) programming to process sensor data and trigger alerts.
- Wi-Fi-based cloud integration using MQTT protocol to send alerts to a cloud dashboard (e.g., ThingSpeak / Adafruit IO / Firebase).
- LED and buzzer-based local alert indicators for immediate on-site notification.

- Simulation of the complete circuit using Wokwi (online IoT simulator).
- Serial monitor logging for debugging and verification of sensor readings.

3.2 Out of Scope

- Camera-based video surveillance or image processing.
- Mobile application development (alerts are sent to a cloud dashboard/email only).
- Physical hardware deployment and installation.
- Multi-zone intrusion detection (single area only in this prototype).

4. Required Components to Develop the Solution

4.1 Hardware Components

Component	Specification / Purpose
ESP32 Microcontroller	Main processing unit with built-in Wi-Fi and Bluetooth (38-pin DevKit v1)
PIR Motion Sensor	HC-SR501 – Detects infrared radiation from human body heat; range: 3–7 m
LED (Red)	Visual alert indicator when motion is detected
Buzzer (Active)	Audio alarm triggered on intrusion detection
Resistor (220Ω)	Current limiting resistor for LED
Jumper Wires	For circuit connections on breadboard
Breadboard	Prototyping platform for connecting components
USB Cable	For programming and powering ESP32

4.2 Software Components

Software / Tool	Purpose
Arduino IDE 2.x	Writing, compiling, and uploading firmware to ESP32
ESP32 Board Package	Arduino core library for ESP32 microcontroller support
PubSubClient Library	MQTT client library for cloud communication
WiFi.h Library	ESP32 Wi-Fi connection management

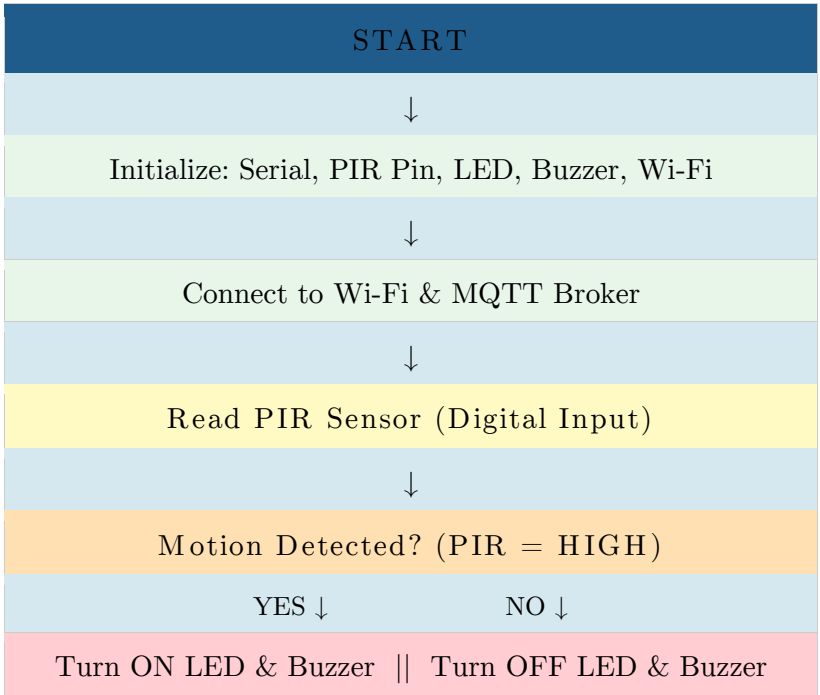
Wokwi Simulator	Online circuit simulation platform (wokwi.com)
VS Code (optional)	Code editor with Wokwi extension for integrated simulation
GitHub	Version control and submission of project files

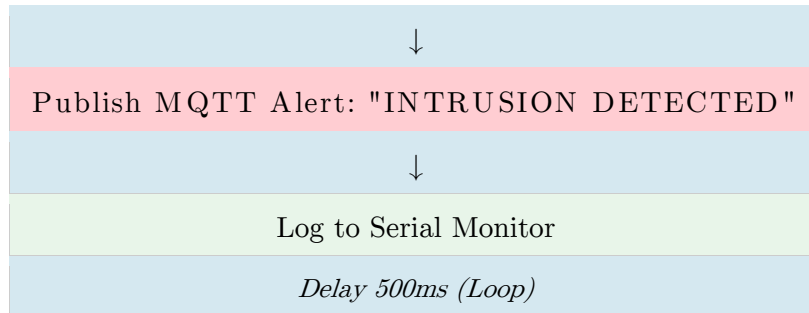
4.3 Cloud Environment

Cloud Platform	Role
Adafruit IO / ThingSpeak	Cloud dashboard to receive and display MQTT alert messages
MQTT Broker (HiveMQ/broker.hivemq.com)	Free public MQTT broker for message routing
Wi-Fi Router / Hotspot	Network connectivity for ESP32 to reach the internet
GitHub Repository	Code and documentation hosting for submission

5. Flowchart of the Code

The following flowchart describes the program logic implemented in the ESP32 firmware:





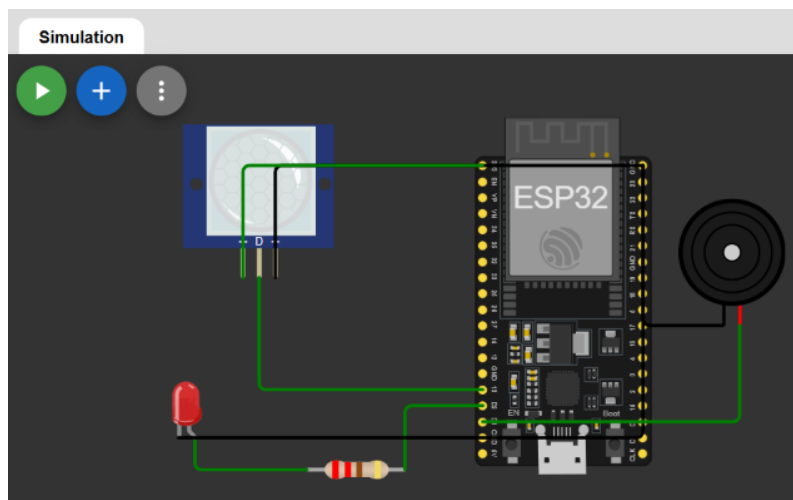
6. Simulated Circuit (Wokwi)

The circuit has been designed and simulated using Wokwi, a free online IoT circuit simulator available at <https://wokwi.com>. The simulation replicates the real-world behavior of the system including PIR sensor triggering, LED and buzzer activation, and serial output.

6.1 Circuit Connections

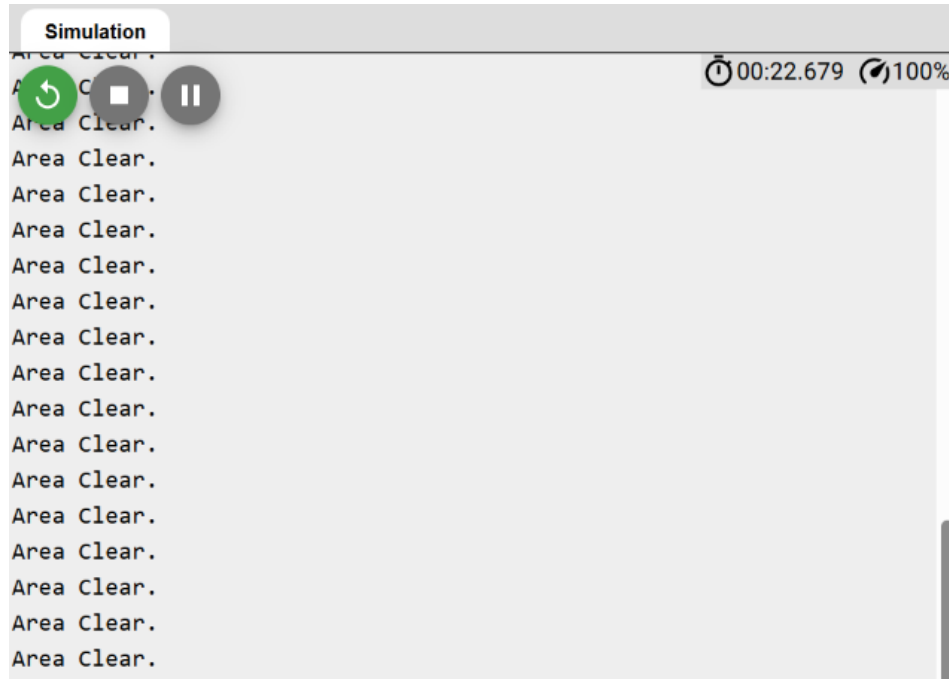
Component	Pin	ESP32 Pin	Notes
PIR (HC-SR501)	VCC	3.3V / 5V	Power supply
PIR (HC-SR501)	OUT	GPIO 13	Digital motion output
PIR (HC-SR501)	GND	GND	Common ground
LED (Red)	Anode (+)	GPIO 2	Via 220Ω resistor
Buzzer (Active)	Positive	GPIO 4	Audio alert output

Circuit Diagram



6.2 Wokwi Simulation Link

Access the simulated circuit at: <https://wokwi.com> (create a new ESP32 project with the code below and add the PIR sensor, LED, and buzzer components as per the connections table above). Simulation Proof:



7. Sample Arduino Code

The following code is uploaded to GitHub and used in the Wokwi simulation:

```
#define PIR_PIN 13
#define LED_PIN 2
#define BUZZER_PIN 3

void setup() {
  Serial.begin(115200);
  pinMode(PIR_PIN, INPUT);
  pinMode(LED_PIN, OUTPUT);
  pinMode(BUZZER_PIN, OUTPUT);
  Serial.println("System Ready. Monitoring...");
}

void loop() {
  int motion = digitalRead(PIR_PIN);
  if (motion == HIGH) {
```

```
digitalWrite(LED_PIN, HIGH);  
digitalWrite(BUZZER_PIN, HIGH);  
Serial.println("ALERT: Motion Detected! Intrusion!");  
} else {  
    digitalWrite(LED_PIN, LOW);  
    digitalWrite(BUZZER_PIN, LOW);  
    Serial.println("Area Clear.");  
}  
delay(500);  
}
```

8. Video of the Demo

A video demonstration of the simulated circuit running on Wokwi has been recorded and uploaded to the GitHub repository. The video covers:

- Starting the Wokwi simulation with the ESP32 and PIR sensor circuit.
- Triggering the PIR sensor to simulate human movement in the restricted area.
- Observing the LED turning ON and the buzzer sounding when motion is detected.
- Viewing the Serial Monitor output showing the alert message.
- Confirming the MQTT message is published to the cloud broker (HiveMQ dashboard or Adafruit IO feed).
- Demonstrating the system resetting when no motion is detected.

The video file is available in the GitHub repository

9. GitHub Repository

All project files including the Arduino code, Wokwi simulation files (diagram.json + sketch.ino), circuit diagram, and this report are hosted at:

<https://github.com/adzzz1609/smart-intrusion-detection>

Repository structure:

- README.md — Project overview and setup instructions
- src/intrusion_detection.ino — Arduino source code
- wokwi/diagram.json — Wokwi circuit layout file
- wokwi/sketch.ino — Code used in Wokwi simulation
- docs/report.docx — This assignment report
- demo/demo_video.mp4 — Video demonstration